

1. Administrative & Study Identification

Full Study Title: The Willow LTE Study With M5049 in Participants With SCLE, DLE and/or SLE **Short Title/Acronym:** WILLOW LTE
Protocol Number: MS200569_0048 **NCT Number:** NCT05540327 **Sponsor Name:** Merck Healthcare KGaA, Darmstadt, Germany (an affiliate of Merck KGaA) **Investigational Product:** Enpatoran (M5049) - TLR7/8 antagonist **Phase of Development:** Phase II **Medical Monitor:** Merck KGaA Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and efficacy of orally administered M5049 (enpatoran) in participants with subacute cutaneous lupus erythematosus (SCLE), discoid lupus erythematosus (DLE), and/or systemic lupus erythematosus (SLE) who have completed the 24-week treatment period of the parent WILLOW study. **Secondary Objectives:** To monitor long-term abnormalities in laboratory parameters and corrected QT intervals.

2.2 Background & Rationale To address the continuous therapeutic needs of patients with SLE and cutaneous lupus manifestations. This Long-Term Extension (LTE) is conducted to further understand the sustained safety, tolerability, and clinical benefits of enpatoran (a TLR7/8 dual antagonist) over an extended treatment period (up to 194 weeks) in patients who successfully completed the initial treatment phases of the parent trial.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Long-term Extension) **Control Type:** Placebo-controlled **Blinding:** Quadruple-blind (Double-blind) **Randomization:** Parallel Assignment (maintaining the parent study allocation)

3.2 Planned Enrollment & Duration Target \$N\$: 379 participants (Actual enrollment, originally estimated at 532) **Number of Sites:** ~123 locations across 22 countries (including US, Europe, Japan, Latin America) **Estimated Study Start:** 16-Sep-2022 **Estimated Primary Completion:** 12-Aug-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Active SCLE, DLE, and/or SLE participants that have successfully completed the 24-week treatment of the WILLOW Study (MS200569_0003). 2. Have a Body Mass Index (BMI) ≤ 40 kg/m² (inclusive) at Screening. 3. Participants who successfully completed Week 48 of Part 1 will have the opportunity to participate in the LTE Part 2.

4.2 Exclusion Criteria 1. Participants who experienced serious event(s) related to the study intervention during the WILLOW study. 2. Ongoing or active clinically significant viral (including SARS-CoV-2), bacterial, or fungal infection, or any major episode of infection requiring hospitalization. 3. Received LTE prohibited medication during the WILLOW study or after the WILLOW study Week 24, or participating in another investigational drug study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Enpatoran (M5049) administered at low dose, medium dose, or high dose, alongside matching placebo, given twice a day (BID). **Route of Administration:** Oral (film-coated tablets) **Duration of Treatment:** Up to 194 weeks of exposure in the extension phase.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care baseline treatments for SLE/CLE maintained from the parent trial. **Prohibited:** Prohibited medications defined in the WILLOW LTE protocol, including concurrent investigational drugs and unapproved immunosuppressive agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	LTE Screening	Up to 4 weeks Transition
consent from WILLOW Week 24/48, Eligibility review, baseline labs			
Baseline (LTE)	Day 0	Continuation of randomized IP	
dosing, Safety Audit	Treatment Visits	Weeks 4 to 194	
Safety Labs, Efficacy Assessments (CLASI, SLEDAI), ECG			
monitoring, IP Dispensation	End of Study	Week 194	Final
Vital Signs, Exit Interview, IP Accountability, Safety Follow-up			

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term Safety Profile as assessed by the incidence of Treatment-Emergent Adverse Events (TEAEs), Serious Adverse Events (SAEs), and Adverse Events of Special Interest (AESIs) during Part 1 and Part 2. **Measurement**

Method: Clinical observation, patient reporting, and Medical Monitor review up to 194 weeks.

7.2 Secondary & Exploratory Endpoints Number of participants with abnormalities in laboratory parameters. Evaluation of QT Interval Corrected over the study period.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient interviews, physical examinations, and laboratory assessments.

Serious Adverse Events (SAEs): Reported within 24 hours of site awareness to the Sponsor safety team. **Data Monitoring Committee (DMC):** External, independent safety data monitoring committee actively reviewing long-term LTE data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations; Intent-to-Treat (ITT) population for long-term efficacy outcomes.

Sample Size Justification: Dependent on the number of rollover participants from the parent WILLOW study (~379 patients).

Handling of Missing Data: Multiple Imputation (MI) or Last Observation Carried Forward (LOCF) for relevant clinical efficacy scales.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard risk-based monitoring with periodic onsite and remote data verification visits. **Audit Trail:** Formal eCRF locking mechanisms and structured data clarification forms handled by the sponsor's data management team.

11. Study Identifiers & Governance

Primary ID: MS200569_0048 **Registry Number:** NCT05540327 **Sponsor/Lead Organization:** Merck Healthcare KGaA **Study Title:** The Willow LTE Study With M5049 in Participants With SCLE, DLE and/or SLE (WILLOW LTE) **Official Regulatory Title:** A Phase II, Double-blind, Dose-Ranging, Parallel, Long-term Extension Study to Evaluate the Safety and Efficacy of Enpatoran in Participants With Subacute Cutaneous Lupus Erythematosus, Discoid Lupus Erythematosus and/or Systemic Lupus Erythematosus Having Completed the WILLOW (MS200569_0003) Study Treatment (WILLOW LTE)

12. Clinical Framework (Lupus Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE), SCLE, DLE
Diagnosis Threshold: Diagnosis of active cutaneous or systemic lupus and successful completion of the WILLOW parent trial treatment
Medical Methodology: Targeted TLR7/8 dual antagonism
Optimization Target: Sustained reduction of lupus disease activity and long-term tolerability

13. Study Procedures & Methodology

Management Pathway: Long-term extension maintenance clinical pathway
Education Protocol (HCP): Protocol training, safety reporting requirements, and blinded dose-ranging management
Education Protocol (Patient): Informed Consent Process for LTE, side-effect awareness, and IP adherence
Quality Audit Mechanism: Electronic patient-reported outcomes (ePRO) and investigator site file audits

14. Observation & Follow-Up Schedule

Total Duration: Up to 194 Follow-up weeks
Onsite Visit Cadence: Regular predefined intervals corresponding to safety and IP dispensation requirements
Remote Monitoring Frequency: Intermittent phone-based check-ins between major clinical visits
Checklist Review Interval: Regular laboratory and ECG reviews for the secondary outcomes

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				